

Figure 1: OpenDaylight architecture (Beryllium)

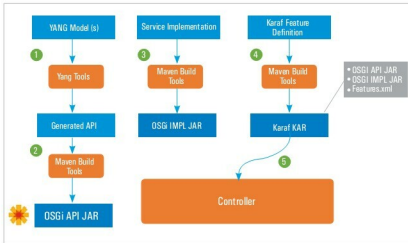


Figure 2: ODL application development workflow

Configuring Maven settings.xml

Execute the following commands to set up your *maven settings.xml* under the *~/.m2* directory:

```
cp -n ~/.m2/settings.xml(, .orig) ; wget -q -O - https://raw.githubusercontent.com/OpenDaylight/odparent/stable/beryllium/ settings.xml > ~/.m2/settings.xml
```

Make sure MAVEN_OPTS and the JAVA_HOME environment variables are set. If not, configure them as part of *~/.bashrc* in Linux, as follows:

```
export MAVEN_OPTS='-Xmx1048m -XX:MaxPermSize=512m'
export JAVA_HOME=/usr/lib/jvm/java-1.7.0-openjdk-amd64
export PATH=$JAVA_HOME/bin:$PATH
```

Setting up the project using Maven Archetype

This is to create a project using the Maven build tool, in which the archetype of 'OpenDaylight-startup-archetype' is used.

```
mvn archetype:generate
-DarchetypeGroupId=org.opendaylight.
controller -DarchetypeArtifactId
=OpenDaylight-startup-archetype
-DarchetypeRepository=https://nexus.
OpenDaylight.org/content/repositories/
public/ -DarchetypeCatalog=https://nexus.
OpenDaylight.org/content/repositories/
public/archetype-catalog.xml
```

This will prompt you for the inputs below:

```
Define value for property 'groupId': : <com.
cloudenablers.networkoptimizer>
Define value for property 'artifactId': :
<networkoptimizer>
[INFO] Using property: version =
1.0.0-SNAPSHOT
Define value for property 'package': com.
cloudenablers.networkoptimizer: :
Define value for property 'classPrefix':
Networkoptimizer: : ${artifactId.
substring(0,1).toUpperCase()}.${artifactId.
substring(1)}
Define value for property 'copyright': :
Copyright (c) 2016 Cloudenablers
```

Once this is completed, it should have generated the *networkoptimizer* directory with its sub-folders in the following hierarchy:

```
├── api
├── artifacts
├── features
├── impl
├── it
├── karaf
└── pom.xml
```

Building, starting and verifying the application

Execute the following command to build the project:

```
mvn clean install
```

Start the OSGi Karaf runtime container, as follows:

```
karaf/target/assembly/bin/karaf
```

Verify that the network optimiser module is built and installed as part of Karaf, as follows:

```
OpenDaylight-user@root>feature:list | grep networkoptimizer
```